Tor Hides Your IP, Not Your Words: Free LLMs as an Attribute-Inference Threat to Users of Network Anonymity Tools

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Abstract.

Tor and VPNs hide a user's IP address, and users often treat that as anonymity. We show that the text such users post remains a strong source of personal information, and that extracting it is now free. On 100 synthetic authors from the SynthPAI dataset, two free-tier Gemini models inferred eight personal attributes from the authors' comments. The best model guessed sex correctly 85.4% of the time, occupation 71.4% and country 72.3%, with a mean of 64.3% over all attributes; most of these attributes are never revealed by an IP address at all. A simple defense, rewriting comments with an LLM before posting, lowered mean accuracy to 49.5% but did not remove any attribute. We release all code and raw model outputs.

1. Introduction

Network anonymity tools such as Tor [1] and VPNs are often described as making their users "anonymous". What they actually protect is the *channel*: they hide which IP address is talking to which server, so a website, an internet provider or an eavesdropper cannot tie traffic to a person's network location. They do nothing to the *content* a user chooses to publish. A comment posted over Tor is just as revealing as the same comment posted from a home connection.

From an information-management point of view, every public post is a record. It persists, it can be collected in bulk, and it carries metadata about its author that the author never meant to disclose: hints about where they live, how old they are, what they do for a living. Contextual-integrity theory [2] frames the harm well: information shared for one purpose (a casual comment about local food) flows into another (a profile of the author) without the author's consent.

Until recently, turning such scattered hints into a profile required a skilled human investigator, which limited the threat to targeted cases. Staab et al. [3] showed that large language models (LLMs) can do this automatically and at scale, with GPT-4 inferring personal attributes from Reddit comments with accuracy close to human investigators at a fraction of the cost and time. Their follow-up dataset, SynthPAI [4], made it possible to study this threat on synthetic, privacy-safe data.

This paper asks what that finding means for the users of anonymity tools specifically, and whether the threat is now available to anyone at no cost. We make three contributions:

1. **Network versus text.** We contrast what a network observer learns from a user's IP address with what an LLM learns from the same user's text, attribute by attribute, and show that removing the IP leaves most of the profile intact.
2. **Zero-cost adversary.** We measure attribute inference using only free-tier models (Google Gemini Flash-class models on the free API tier), showing that the attack no longer requires a paid frontier model.
3. **A simple defense, measured.** We test a "rewrite before posting" step, in which an LLM paraphrases comments to strip personal clues, and measure how much it reduces inference accuracy and how much meaning it keeps.

The code and all raw model outputs are released so that every number in this paper can be recomputed.

2. Threat Model

We consider a user who posts short public comments (for example on a forum) while connected through Tor or a VPN. The user's goal is to stay unlinkable to their real-world identity.

The **network adversary** observes connections to the forum. Without an anonymity tool, the source IP address can be geolocated, which usually reveals the country and sometimes the city, though city-level geolocation databases are often wrong [5]. It reveals nothing about age, sex, education, occupation, income or relationship status. With Tor or a VPN, the adversary sees only the exit relay's address, which reveals nothing about the user.

The **text adversary** has no network access at all. It collects the user's public comments (by username) and sends them to an LLM with a prompt asking for a best guess of eight personal attributes. It needs no technical skill, no training data and, as we show, no money. Its output is a structured profile that could be combined with other sources to narrow down or re-identify the user.

Anonymity tools defeat the first adversary completely and the second not at all. The question is how much the second one learns.

3. Related Work

Authorship and stylometry. Linking anonymous text to its author by writing style is a long-studied threat; Narayanan et al. [6] showed author identification at internet scale with classical machine learning. That line of work identifies *which* known author wrote a text; attribute inference instead describes an unknown author.

LLM attribute inference. Staab et al. [3] introduced the PersonalReddit evaluation and showed that GPT-4 infers location, age, sex, income and other attributes from real Reddit comments with high accuracy. SynthPAI [4] provides 7,800+ synthetic comments from 300 synthetic profiles with human labels, and showed that conclusions drawn on it match those drawn on real data across 18 models. We use SynthPAI and adapt its inference setup.

Defenses. Staab et al. [7] showed that LLM-based adversarial anonymization outperforms traditional text anonymizers, using an iterative loop in which an attacker model guides the rewriter. We test a much simpler, single-pass rewrite that a privacy tool could run locally before posting, and report how far that simple step gets.

Our work differs from these in framing (the specific gap between network and text anonymity), in the adversary (free-tier models only) and in measuring a deployable one-shot defense.

4. Method

Data. We use SynthPAI [4] (CC BY-NC-SA 4.0). Each of its 300 synthetic authors has a ground-truth profile with eight attributes, and each comment carries human-reviewer labels marking which attributes a human could infer from it. We sample 100 authors (fixed seed) among those with at least one human-inferable attribute, concatenate each author's comments in dataset order and truncate them to 3,000 characters. For each author we score only the attributes that a human reviewer marked as inferable, so the task is never impossible by construction.

Models. The adversary uses only models available on the free tier of the Google Gemini API at the time of the run (2026-10-18): `gemini-3.1-flash-lite` and `gemini-3.5-flash-lite`. All calls use temperature 0.

Inference prompt. Adapted from Staab et al. [3]: the model is told it is an expert investigator, is given the comments, is asked to reason about every clue and is required to return a JSON object with a guess (or null) for each attribute, with fixed answer options for sex, education level, income level and relationship status.

Defense. Before inference, `gemini-3.1-flash-lite` rewrites each author's comments with the instruction to keep meaning and tone but remove any clue about the eight attributes. We then run the same inference on the rewritten text. To check that the rewrite still says the same thing, the same model rates meaning preservation from 1 to 5 for the first 30 authors.

Scoring. Age is correct within ±5 years. Sex, income level, education level and relationship status are compared after normalization. City, birthplace and occupation are free text, so `gemini-3.1-flash-lite` acts as a strict yes/no judge ("same city?", "same or very similar occupation?"). We additionally report **country** accuracy (exact country match) because country is what IP geolocation reveals. Empty or unparseable answers count as wrong. Requests that failed because of free-tier quota limits returned no answer at all, so they are excluded and their number is reported. We report accuracy with 95% bootstrap confidence intervals (1,000 resamples).

5. Results

Table 1 gives inference accuracy for every attribute, model and condition, and Figure 1 shows the same numbers with their 95% confidence intervals. Of the 400 inference requests, 41 were refused by free-tier quota limits and are excluded; judge calls refused the same way are also excluded, which is why the number of scored authors differs slightly between cells.

Attribute	**n**	**`gemini-3.1-flash-lite`**		**`gemini-3.5-flash-lite`**	
		orig.	rewr.	orig.	rewr.
Age (±5 y)	54	68.5%	54.6%	66%	43.4%
Sex	41	85.4%	78.6%	85.4%	70.7%
City	47	46.8%	25.5%	45%	15.8%
Country	47	72.3%	44.7%	54.4%	23.9%
Birthplace	10	40%	10%	28.6%	10%
Education	51	70.6%	65.4%	60%	42%
Occupation	77	71.4%	52%	71.4%	54.2%
Income level	57	70.2%	69%	57.9%	59.7%
Relationship	45	53.3%	43.5%	43.2%	25%

Table 1: Attribute-inference accuracy on original and rewritten comments. n is the number of scored authors for the first model on original text (authors for whom a human marked the attribute as inferable, minus refused requests); per-cell counts are in the released CSV.

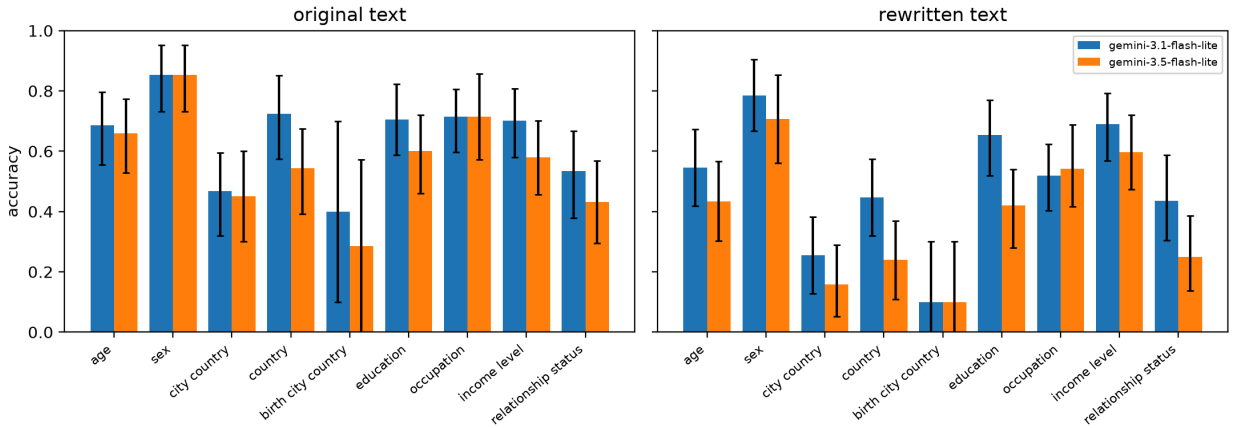


Figure 1: Accuracy per attribute with 95% bootstrap confidence intervals, before (left) and after (right) the rewrite defense.

Network versus text. Table 2 puts the two adversaries side by side. Without an anonymity tool, the network adversary learns the country and, unreliably, the city from the IP address; with Tor it learns nothing. The text adversary, which is unaffected by Tor, recovers every attribute well above zero. Averaged over the attributes, the best free model is right 64.3% of the time on the original comments.

Attribute	IP, no Tor	IP, Tor	Text	Text, rewritten
Age (±5 y)	none	none	68.5%	54.6%
Sex	none	none	85.4%	78.6%
City	often (IP geolocation, city level is unreliable)	none	46.8%	25.5%
Country	yes (IP geolocation)	none	72.3%	44.7%
Birthplace	none	none	40%	10%
Education	none	none	70.6%	65.4%
Occupation	none	none	71.4%	54.2%
Income level	none	none	70.2%	69%
Relationship	none	none	53.3%	43.5%

Table 2: What each adversary learns. Network columns are analytical (from the threat model, [5]); text columns are the best measured model accuracy.

Rewrite defense. The single-pass rewrite lowers the best model's mean accuracy from 64.3% to 49.5%, but no attribute falls to zero. The rewriter rated its own rewrites at 5 out of 5 for meaning preservation on average (30 authors).

6. Discussion

Anonymity is a property of the whole record. Information-management practice treats a record as data plus metadata plus context. Anonymity tools strip one piece of metadata, the network address, and leave the record itself untouched. Our results show that the remaining record still carries most of a personal profile, and that reading it no longer needs an expert: a free model, a published prompt and a few seconds per author are enough. For an anonymity-tool user, the network address is often not the most revealing thing they send.

Why the rewrite falls short. Reading the rewritten comments shows why a single pass leaks. The rewriter removes explicit statements ("I'm a nurse in Montreal") but keeps the experiences that imply them, because those experiences *are* the meaning it was told to preserve. One rewritten author still "zone[s] into coding mode" and enjoys "crushing bugs"; another describes "grading marathons" in "the break room" and the responsibility of "teaching students"; a third complains about paychecks "feeling thin before loan payments". Each line is harmless alone, but together they let the model guess occupation, age and income. Meaning preservation and attribute removal pull in opposite directions, so a defense that keeps the conversation useful will rarely remove everything.

Implications for tool builders. Tools such as Tor-based VPN clients could offer an optional local step that warns users before posting, for example by running a small model over the draft and listing the attributes it can infer. Showing users what their own words reveal may protect them more than silently rewriting their text, and it keeps the user in control of what is published.

7. Limitations

Synthetic data. SynthPAI comments are generated, not written by real people. Its authors report that conclusions carry over to real data [4], but real users may give fewer or different clues.

Sample size. We evaluate 100 authors, and each attribute is scored only for authors where a human marked it inferable, so some cells rest on a few dozen authors. The confidence intervals in Figure 1 are wide accordingly.

One vendor, two models. Both models are Gemini Flash-Lite models from one provider. Other free models, or larger paid ones, may be stronger or weaker; ours is a lower bound on what a determined adversary can do for free. A planned third model (Gemini 3.5 Flash) was dropped because free-tier rate limits made it too slow to complete.

Quota limits. The free tier refused 41 of 400 inference requests and some judge calls. We exclude these rather than count them as wrong, which assumes the refusals are unrelated to how revealing the text is (quota refusals depend on request timing, not content).

Free models change. Free-tier models are updated and withdrawn without notice. We release every raw request and response so the results can be checked even if the models disappear.

LLM judge and self-rating. Free-text attributes are scored by an LLM judge, and meaning preservation is rated by the same model that did the rewriting, which may be lenient toward its own output. The judge was not validated against human ratings in this study; its decisions are released for inspection.

One rewrite prompt. We test a single, simple rewrite prompt. Stronger, iterative anonymizers [7] would likely do better; our result shows what a minimal step achieves, not the limit of defenses.

Network adversary not measured. The network columns of Table 2 come from the threat model and prior work, not from our own measurement.

8. Ethics

All profiles and comments are synthetic; no real person's data was collected or inferred. The attack we describe is already public [3], and our contribution is to measure its cost and its relevance to anonymity-tool users so that those users and tool builders can respond. We release code and outputs to support defenses, not new attack capabilities: the prompt is a lightly adapted version of a published one.

9. Conclusion

Tor and VPNs hide where a user connects from, not what the user says. Using only free-tier models and no technical effort, an adversary in our study inferred personal attributes from comments with a mean best-model accuracy of 64.3%, covering attributes that an IP address never reveals. A single rewrite before posting reduced this to 49.5% but left every attribute partly exposed. Anonymity tools that care about their users' privacy should treat text as part of the attack surface, for example by offering a local "rewrite before posting" step and warning users that their words, not only their address, identify them. Code, prompts and all raw model outputs are available for anyone to rerun or extend.

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